Aviv Shaked

Cybersecurity Developer & Researcher · Haifa, Israel

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SUMMARY

C++20 cybersecurity developer and researcher specializing in network security, binary exploitation, and reverse engineering. Graduate of **Magshimim+**, Israel's selectively admitted national cybersecurity track, consistently self-driven beyond the curriculum and proven to deliver organized, professional work in both team and independent settings. Co-developed **HitchHiker**, a production covert communication platform embedding encrypted messages into live traffic with zero new packets; placed **2nd globally** co-leading a team in AI4Y (2023).

TECHNICAL SKILLS

Languages	C++20 (primary, 3+ years), Python, C, x86/x86-64 Assembly, SQL, C#
C++ & Systems	RAII, smart pointers, multi-threading, modern C++20 idioms, CMake, interface abstraction, modular builds
Security	Binary exploitation, ROP chains, buffer overflows, ASLR bypass, reverse engineering, network steganography, protocol analysis, threat modeling, exploit development, HackTheBox (RE, Crypto, Pwn)
Networking	Full network stack (L2-L7): TCP/IP, UDP, DNS, TLS, ICMP, ARP, DHCP, header-level field manipulation, kernel-level packet interception, iptables, Wireshark, Scapy
Cryptography	Asymmetric and symmetric encryption, key exchange, key derivation, password hashing and salting, cryptanalysis, hash attacks, applied in production (Crypto++)
AI / ML	Neural network fundamentals, LLM/API integration, Python AI/ML tooling (PyTorch, TensorFlow, Scikit-learn)
Tools	Binary Ninja, IDA Pro, GDB, nmap, Git, Kali Linux, Docker, AWS EC2, OpenVPN, JetBrains IDEs

PROJECTS

HitchHiker: Network Steganography Platform

C++20 · 2025-2026 · Completed

Co-developed with Daniel Varkin · Source available on request · Architecture: patcholie.github.io

C++20 | Crypto++ | NFQueue | FTXUI | Redis | SQLite | Docker | AWS EC2 | CMake

- Co-architected a covert communication platform that embeds encrypted messages into live network traffic: **zero new packets generated**, no observable network presence, undetectable by design
- Full applied cryptography stack (key exchange, symmetric encryption, key derivation, password hashing) applied at bit-field precision across multiple protocol layers via kernel-level packet interception
- **Pluggable carrier architecture** behind a unified HitchHikerAPI; three independent CMake targets (lib / server / client), Redis and SQLite storage, full TUI client (FTXUI); server deployed on AWS EC2, clients run on Linux via Docker

Multiplayer Trivia Game System

C++ / C# / Unity · Magshimim · 2024

C++ | C# | Unity | SQLite | TCP Sockets

- Designed a custom application-layer protocol from scratch; concurrent TCP server, C# Unity client, and SQLite database integrated end-to-end with clean OOP architecture; scored 130/100 including bonus marks

Analisis: AI Wound Treatment Platform

Python · 2023 · Public

- Co-led development of an AI-powered wound diagnosis and treatment guidance platform; **placed 2nd globally** in the AI4Y World Competition (2023) across international teams

Mancala-AI

Python · PyTorch · 2025-2026

- Multi-agent AI framework with four strategy types including a neural network agent trained via self-play reinforcement learning; full training and tournament evaluation pipeline

EXPERIENCE

Python Automation Developer, AllBikes

2022-2023 · Remote

- Built Python automation tools for web scraping and product catalog structuring; **tooling adopted by the full team**, delivered independently in a remote professional environment while in secondary school

EDUCATION

Magshimim Cyber, Advanced Track (Magshimim+)

2023-2026

Israel's elite national cybersecurity program, selective advanced track. Curriculum: networking, binary exploitation, reverse engineering, cryptography, systems programming (C/C++), Linux, team-based security development

BOSMAT, Haifa

2020-2026

Computer Science and AI, 10 study units (maximum track)

AWARDS & RECOGNITION

2nd Place, AI4Y World Competition · 2023 · Global

Magshimim+ Advanced Track · Graduate 2026 **Magshimim Cyber** · Graduate 2026